

Planning Context

The Region and Its Demographics

The Upper Ohio River Corridor which is the context of this plan and encompasses the four communities to be discussed is located in western Pennsylvania northwest of Pittsburgh along the 40 miles of the Ohio River which travel from the City of Pittsburgh into Ohio. This corridor is divided by Allegheny and Beaver Counties which are then subdivided by hundreds of smaller municipalities. The municipalities which are at the center of this plan listed from west to east are the Borough of Midland, the Borough of Monaca, the City of Aliquippa and Coraopolis Borough.

Allegheny County is 730 square miles in size with 1,675 persons per square mile and is comprised of 130 municipalities including the City of Pittsburgh (which is also the County seat) as well as its surrounding suburbs. In 2011 the County’s population was 1.22 million up 3% from 2010. Demographically, the County is 81.8% White, 13.3% Black and 2.9% Asian with a median household income of \$47,961. In lieu of recent positive population growth trends Allegheny County, like many other communities in the region, has had a declining population each decade from 1970 to 2000. The recent positive growth stems from a continuing and

focused economic restructuring of the region, redirecting a manufacturing workforce into various alternative professions such as tech and medical industries.

Beaver County is smaller than Allegheny County in size, only 434 square miles, and less densely inhabited with 392 persons per square mile which makes sense considering it does not contain an international city such as Pittsburgh in Allegheny County. Its communities and small towns have a more residential character with brick paved streets and fenced backyards.



The Upper Ohio River from its confluence in Pittsburgh to the Ohio state line

In 2011 its population of 170,414 was down by 1% from 2010 and is 91.3% White, 6.4% Black and 0.5% Asian with a median household income of \$46,190. This decline in population is something that has been occurring in the County since the 1970s and is primarily caused by outmigration directly linked to the regional decline in the Steel Industry. The population that has remained is aging. The median age as of 2011 was 40.7 years old, above the national average of 36.8.

Upper Ohio River Corridor

General Overview

The Upper Ohio River is a designation given here to describe the portion of the Ohio River located inside of Pennsylvania. This section stretches 40.0 river miles in length and flows in a northwesterly direction from Pittsburgh to the Ohio state line. A general discussion of the entire Ohio River can be found in the Appendix portion of this plan.

The floodplain is very constricted in this section, with the river at ordinary high water typically accounting for 30 to 60 percent of total floodplain area (defined as approximately level terrain between steeper bluffs). Low and high-density urban uses (commercial, industrial, high and low-density residential) dominate the floodplain, including islands in the upper half of the river reach near to Pittsburgh. Deciduous forest dominates bluffs or steep areas adjacent to the river outside urbanized areas, whereas low-density residential or deciduous forest dominates bluffs within the urbanized areas. Low and high-density urban uses exist in upland areas in the east (Pittsburgh area) and also in Beaver, PA, and East Liverpool, Ohio. Deciduous forest and pasture are the principal land uses outside of the urbanized areas. Only a few small areas of cropland or mixed forest (mixed deciduous and evergreen forest) are visible.

Five islands are located in this stretch of river, two of which are designated as Ohio River Islands National Wildlife Refuges by the Nature Conservancy. These two islands are George Island and Phillis Island, both located in Beaver County near Midland, 33 miles along the Ohio River outside of Pittsburgh.

Water Quality

One of the major threats to this section of the Ohio River is acid mine drainage. Upstream mines along the Monongahela River deposit high levels of acidic material into the water system. Eventually, this material flows into the Ohio River. From 1945 to 1960, at the Pennsylvania, Ohio, West Virginia state line, a net average acid load of about 350 tons per day moved along the river. This influx of heavily acidic material shifts the pH of the water from what is generally around 7 to higher acidity levels. In extreme cases these levels can reach a pH of -3.6. This is highly detrimental to the health of impacted ecosystems and can create a “yellowing” of local streams and water systems. This yellowing is actually the deposition of ions such as iron oxides and oxyhydroxides.

Conclusions: While improvements in mine standards and several cleanup efforts have made this issue less destructive, acid mine drainage still impacts the area. The Upper Ohio River inherits many of these issues from upriver

sources so what can be done to mitigate the issue within the study area is restricted to reactionary measures. That being said several reactionary measures have proved to effective in treating water quality issues. One way to address the issue is by adding more basic material to water such as limestone or by treating the water in lime silo holding tanks before releasing it back into the river system. Vegetation can also help mediate acid mine drainage issues in the form of constructed wetlands projects. These wetlands help to capture some of the metals flowing through the contaminated water.

Landscape Characteristics

In Pennsylvania, the Ohio River flows within the unglaciated part of the Appalachian Plateau. Chartiers and Raccoon Creeks, the major tributaries entering from the south, drain a hilly section of the plateau. Between the Pennsylvania border and Wheeling, West Virginia, the river continues to flow though an area unmodified by glacial action, so the topography remains rather rugged and dramatic. It is drained by many short streams entrenched in fairly deep, narrow valleys.

Habitat Conditions

Mussel habitats are some of the most disturbed habitats in the Upper Ohio River. Mussels were completely extirpated from the upper portion of the Ohio River during the early 20th Century. Arnold Ortmann (1919), in his classic monograph of the Naiades of Pennsylvania, reported the total demise of the mussel fauna of many tributaries of the Upper Ohio River, mostly from the impacts of acid drainage from bituminous coal mining, however, in 1919 he was still able to document either the presence, or relatively recent occurrence, of a diverse assemblage of at least 39 species of mussels in the the Upper Ohio River.

The degree, extent, and duration to which previous water quality conditions adversely affected survival of mussels along the Upper Ohio River cannot be precisely determined from historical data. However, as early as 1914, the U.S. Public Health Service documented episodic acidification of the Ohio River extending as far as 170 miles downstream of Pittsburgh (U.S. Public Health Service 1924). Data collected by the U.S. Geological Survey and the Pennsylvania Department of Forests and Waters provide a good picture of Ohio River water quality conditions (at a station 17 miles downstream of Pittsburgh) during the late 1940s. During this time the pH of the Ohio River ranged from 3.65 to 6.6 and averaged 5.3 (pH of 7.0 is neutral). Other water quality factors affecting mussels in the upper river included low DO, high turbidity and suspended solids, and high concentrations of contaminants (e.g., metals and organic compounds).

Researchers have recently found evidence of 54 species, including 35 species that historically appeared in this portion of river and 19 newly established species. This considerably expands the number of species reported for the Upper Ohio River in 1989. Of the 54 species for which evidence was found (including weathered and sub fossil shells), one species is presumed extinct, eight species are considered extirpated from the study area and the status of two species is unknown. Ten species thought extirpated from the upper river, however, were found alive or as freshly dead shells. Indications of recent recruitment were observed for 15 species, including five species previously thought extirpated.

Several mammals also inhabit this portion of the Ohio River. Beavers, muskrats and river otters greatly impact this ecosystem by building dams and burrowing in river banks. At times these animals and their interventions help the ecosystem. Beaver dams which can be as tall as ten feet in height help to filter water but can also plug up narrow portions of the river or its feeder streams. These dams are often passed down as habitats from the beaver who construct them to muskrats and river otters.

All of this wildlife habitat is made possible by the plant communities that line the Ohio River. These communities provide material from which habitats can be created (such as the beaver’s dam) but also make the banks of the river hospitable by rooting in to prevent soil erosion. Without such vegetation the banks of the river would be washed out and unable to foster the animal communities that they currently do. River edge plants such as cattails, rushes, and sedges help to stabilizing these banks while other plants such as Wild Celery remain completely submersed in water. Near the river in forested floodplain areas flood tolerant trees such as Eastern Cottonwood, Swamp White Oak, American Sycamore, and Black Willow grow.

Conclusions: Supporting these native vegetation communities is one of the best ways to protect and support the habitats of native wildlife as the two exist in symbiosis with one another. The design and construction of wetlands systems comprised of native plants illustrates one method of fostering this relationship. This system could also help to negate some of the issues associated with acid mine drainage.

Maritime Activity and Related Waterborne Industry

Nautical traffic through the Upper Ohio River totaled 24 million tons in 2006, with coal accounting for 17 million tons or 74 percent of the total traffic. Coal moves both upstream and downstream depending on the characteristics of the coal, and on the locations of mines and coal consuming facilities.

Electric utilities move coal from mines in Monongahela Basin upstream of the Upper Ohio River to generating plants downstream on the Ohio

while steel and other electric generating companies move coal from mines downstream of the targeted brownfields sites to West Virginia and Kentucky mines to coking and generating plants upstream in the Monongahela Basin. Construction companies also use the river to move materials like stone, sand and gravel, and cement into the Pittsburgh area. The estimated transportation savings attributable to the Upper Ohio River subsystem is \$144 million annually.

Supporting Infrastructure: Concerns have been raised over the reliability of this river system as a navigation route, especially as many of the locks and dams begin to age. Emsworth, Dashields and Montgomery Locks and Dams have been in operation for 79, 71 and 64 years, respectively, and are nearing the end of their life expectancy. Built prior to air-entrained concrete, all three structures experience structural condition fatigue of the concrete causing unreliable structural capacity of the lock walls. Also, none of the structures meet modern design criteria.

The Emsworth locks have been operational since 1922 while the original dams were rebuilt into gated structures and have been operational since 1938. Dashields and Montgomery Locks and Dams have been in operation since 1929 and 1936, respectively. Several issues at these facilities were temporarily addressed during major rehabilitation projects at Emsworth, Dashields and Montgomery Locks and Dams undertaken in the 1980s and early 1990s. These efforts were only intended as stop-gap measures designed to extend the useful life of the structures for an additional 25 years. Given the effects of weathering and age on the projects, there were concerns over their continued reliability. Anchoring and other temporary

solutions were implemented in an attempt to extend their life until more permanent repairs could be evaluated and implemented. Thus, the rehabilitation program was kept limited in scope and was guided by expediency in lieu of lock closure time constraints, costs or feasibility.

Many known problems were only partially addressed; others were left untouched because repairs were found to be non-feasible, while still others were left uncorrected because they were considered not critical. As an example, deteriorated concrete was left in place instead of being removed and was mostly covered over. Thus, while the surface appearance of the locks, particularly the top of the lock walls, appears in good condition, hidden below that surface are most of the problems that existed before the major rehabilitation efforts.

Conclusions: Inadequate lock capacity is an issue facing the productivity of the Upper Ohio River. This issue, along with deteriorating dams conditions along the corridor will be evaluated in future feasibility studies conducted by the US Army Corps of Engineers, however, it is safe to assume that regardless of these findings the lock and dam systems in place will not change dramatically as it is crucial to the economic success of the corridor.

The Ohio River in Relation to Study Areas

General Overview

All of the municipalities within the Area-Wide Brownfields Planning Project fall into the watershed designated by the EPA as the Upper Ohio Watershed. The United States Geological Survey (USGS) cataloging number for the area is 05030101. Knowing this cataloging number and using it to search the USGS database will provide additional information about the area not included in this report. In addition to the municipalities mentioned and the surrounding areas in Pennsylvania, the watershed area also stretches into West Virginia and Ohio further downstream.

Some of the streams running through municipalities near the sites in question are as follows: Monaca – Elkhorn Run; Aliquippa – Logstown Run; Coraopolis – Moon Run, Montour Run, McCabe Run.

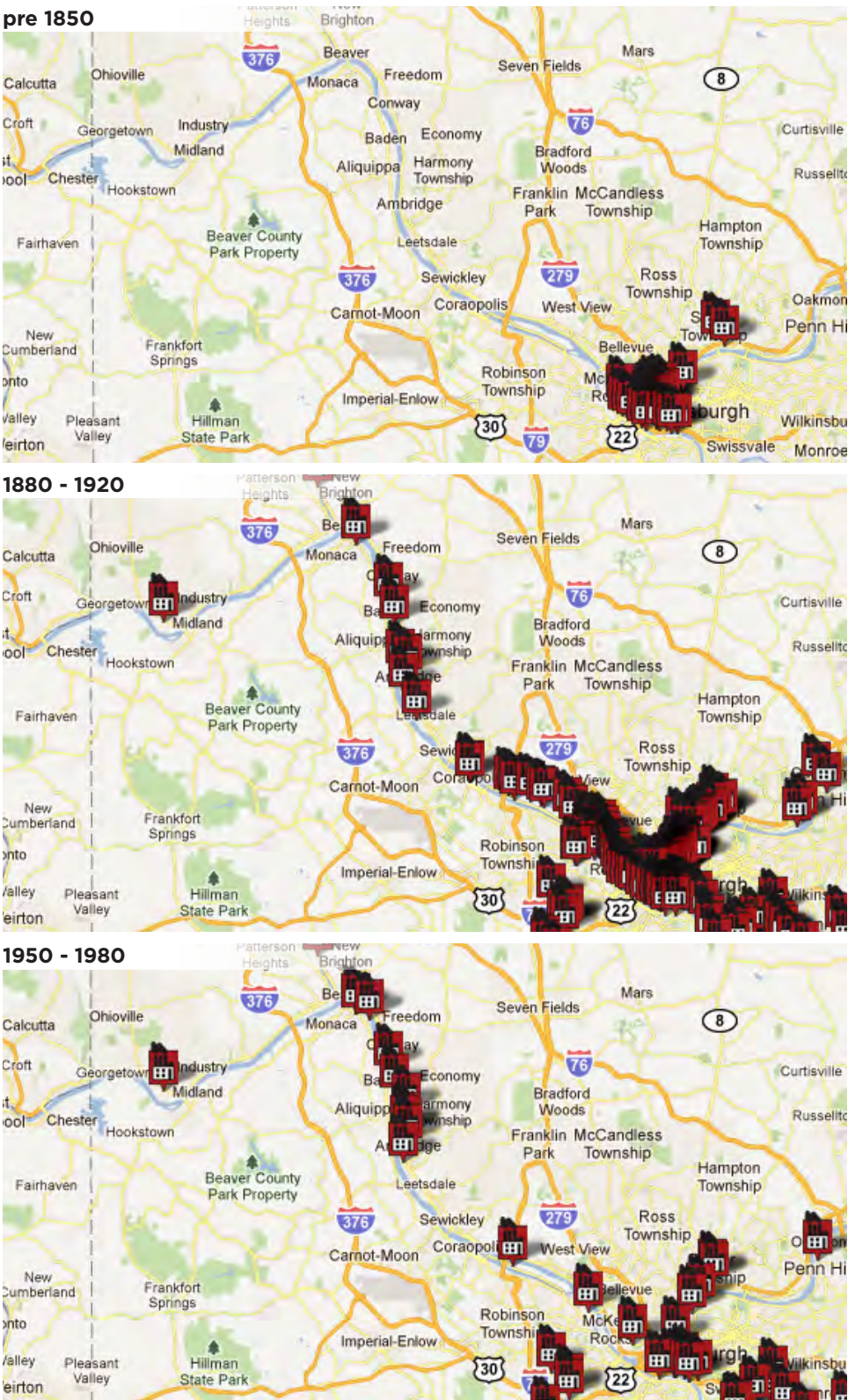
Landscape Characteristics

Land Use Condition: Both north and south banks of the Ohio River have substantial industrial/commercial uses. An active CSX rail corridor runs along the south bank of the river until it reaches Monaca where it crosses via dedicated rail bridge to the north bank. From there (Bridgewater) it continues along the Beaver River eventually crossing into Ohio. Also from Bridgewater, a Norfolk Southern rail line travels along the north shore of the Ohio River to the Ohio State line.



The Upper Ohio River from Monaca

Area-Wide Approach



This series of maps from the Rivers of Steel National Heritage Area shows the increase and decline of steel factories along the Ohio River and its tributaries from 1850 to 2010

Bank Vegetation Character: Upper Ohio River north banks are often steep and over twenty feet in elevation. The canopy ranges from intermittent to continuous. Upper Ohio River south banks in this section are also steep as a result of shelves created for the railroad rights-of-way. Slopes decrease in intensity towards Pittsburgh. Near Brunot Island they average around 15 – 20 feet in height.

Public Access and River Use: Public river access within the portion of the Ohio River in the Area-Wide Brownfields Planning Project is mixed in ownership and relatively sparse. Industries still make use of the river with barge slips in Monaca and Midland. Alternative access is limited to a few privately owned yacht clubs such as the one in Coraopolis and privately owned boat slips attached to residential properties such as those in Monaca.

River Edge Conditions

The edge of the Ohio River is defined here as the space where the river contacts dry land. It is the pivotal transitional space where these two elements (land and water) converge and overlap as part of the ongoing swell and recess natural to any river system. Because so many things come together here the proper management of the edge condition is crucial to the health of both the river and the surrounding landscape. Without proper consideration this transitional zone could become degraded polluting the river with sediment and debris while scarring the land and creating unusable portions of what could otherwise be highly valuable land. In the Pittsburgh region, the importance of the river’s edge is well understood and has been the focus of recent studies dedicated to better understanding and maintaining the health of this edge condition.

The *Three Rivers Park Landscape Management Guidelines* report prepared by Andropogon and Associates Ltd. for Riverlife Task Force provides a comprehensive look at river edge conditions in and around the City of Pittsburgh, Pennsylvania. The report focuses on the Ohio, Allegheny and Monongahela Rivers examining them from their confluence in downtown Pittsburgh to various distances beyond. A three mile stretch of the Ohio River is analyzed from Point State Park in Pittsburgh to Chartiers Creek downstream. The report goes beyond the analysis of what *is* and begins to propose what *should be*. Design guidelines are then provided helping to demonstrate how this transition might come to fruition.

The issues facing the health of the Ohio River in the scope of the *Three Rivers Park Landscape Management Guidelines* report are primarily man-made. For instance, though river flooding is a natural process, it has been both increased and constrained by human interventions. Impervious surfaces such as roads, parking lots and buildings impede rainwater’s ability to be absorbed by the ground. In lieu of this absorption, increased rainwater flows at a higher volume and velocity into the adjacent river,

collecting street debris and contaminants along the way. In an effort to mediate this increased flooding the riverbanks have been “armored” with gabions, concrete and stone retention walls. These manipulations along the river have made it possible for invasive plant communities to flourish. These communities thrive on distressed conditions where native plants have been cleared for construction projects. In other instances, native vegetation has been cleared to improve views of the rivers. Invasive plant communities spread quickly and result in “decreased species richness, decreased biodiversity and less resilient ecosystems.”

To combat these issues several recommendations have been made. Some of the suggestions are policy based while others take a more shovel ready approach. One such policy based initiative was a suggested amendment to municipal zoning ordinances defining a riverfront overlay district. This district would adhere to more specific landscape guidelines such as the planting of native plant species in an effort to foster diversity and a healthy river ecosystem. A more shovel ready approach called for replacing inland paved surfaces with pervious surfaces such as lawns, street trees and hedgerows. All of these recommendations were geared towards improving the health of the Ohio River ecosystem while making it an asset to the local community.

Conclusions: Further downstream along the Ohio River many similar edge conditions exist. As such, many of the recommendations outlined in the *Three Rivers Park Landscape Management Guidelines* report remain relevant for all of the ORBCo communities. Although this downstream portion of the river (from Coraopolis to the Ohio state line) is less urbanized than the portion nearer to Pittsburgh, it has been heavily industrialized resulting in many similar issues. For instance, the impervious surfaces of industrial sites quickly shed stormwater and deposit the water over steep embankments into local streams and the adjacent Ohio River. These steep embankments are often home to invasive plant communities and man-made interventions such as docks and boat slips for barges. Additionally, this downstream portion of the Ohio River has to deal with what takes place upstream closer to Pittsburgh such as increased flooding from impervious surfaces as well as any pathogens that may have gotten into the water. Detailed *Upper Ohio River Edge Mapping* can be found in the Appendix portion of this plan.

Moving forward, the *Three Rivers Park Landscape Management Guidelines* and strategies it proposes should be used as a base framework to address several of the complicated issues facing the Upper Ohio River, however, new recommendations should be investigated to address several characteristics unique to the ORBCo communities. Acid mine drainage, an active industrial base and large portions of the river’s edge sloping at greater than 25% represent a few of these challenging characteristics. An extended set of recommendations would insure that the river’s ecosystem has the potential to become as healthy as possible while also providing an amenity to the local communities.

General Upper Ohio River Corridor Information

- 40.0 river miles in length.
- Travels from Allegheny County through Beaver County.
- Contains three Locks and Dams: Emsworth, Dashields and Montgomery.
- Contains five islands including two designated as National Wildlife Refuges.

